

A controlled path to virtualised pharma IT.

PI Health Sciences Limited

HPE datacenter & VMware implementation

Two HPE hosts, VMware management and separated network services support the pharma IT design. This case study explains legacy migration, controlled cutover and daily operations.



2

HPE RACK SERVERS

32

VMWARE CORE UNITS

5

INSTALLATION MAN-DAYS

COMPANY FACILITY | CLIENT-SUPPLIED PHOTOGRAPH

TECHNICAL PROJECT VIEW

Architecture • Core configuration • Migration

Native 2D diagrams and 3D rack art. Design values are illustrative where site configuration records were not supplied.

Project basis | Hardware, software and the client requirement

The modernisation requirement is to move legacy server services into a manageable VMware environment and organise core-network connectivity around approved application, administration and backup paths. The orders establish the compute and licence package; the migration and core configuration below translate the user’s requirement into a technical delivery design.

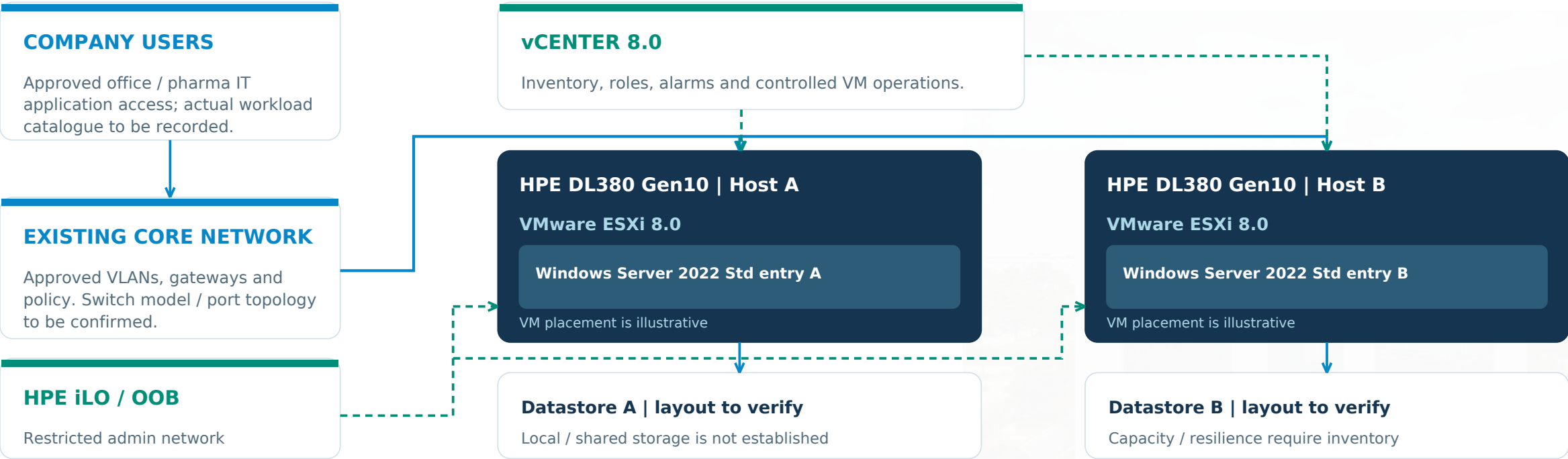
Package	Source / quantity	Technical interpretation
COMPUTE	PO 5100006035 22 March 2025 2 x HPE ProLiant DL380 Gen10 8SFF	Two physical rack servers. CPU, RAM, populated drives, controller and NIC configuration are not detailed in the PO.
WINDOWS LICENSING	PO 5100006083 26 March 2025 2 x Windows Server 2025 Datacenter 16-core	Licence purchase differs from the Windows Server 2022 Standard entries in the access confirmation; verify downgrade rights, keys and physical-core coverage.
VMWARE / SERVICE	VCF-VSP-FND-8 32 core units Installation 5 man-days	VMware vSphere Foundation 8 line and installation effort. The separate image lists 32-core pricing for years 1, 2 and 3; validate activated entitlement dates.
ACCESS CHECK	Client message: working credentials 2 HPE management + 2 ESXi + 2 Windows + 1 vCenter entries	Access to the listed systems is confirmed. This is not a performance test, application acceptance or migration completion certificate.

READ THE IMPLEMENTATION

03 Architecture | 04 Core network | 05 Legacy migration | 06 HPE / VMware build | 07 Pharma operations | 08 Verification | 09 Client & support

Architecture | Compute, applications and management

Business traffic reaches the VMs through the approved core network. vCenter coordinates host and VM administration; HPE iLO provides separate hardware management. The sketch maps the supplied platform entries and shows the intended management boundaries.



Storage and host clustering must be verified before enabling HA or live migration. Two hosts alone do not establish automatic failover.
SOLID BLUE = business / storage connection | DASHED GREEN = management path | Logical VM placement only; no actual IPs or port numbers supplied.

Core network | VLANs, uplinks and controlled access

Build a consistent port-group map on both hosts, then match it to the physical core ports. Keep management services restricted and allow only the application flows approved by the company. The IDs below are examples, not the site’s recorded configuration.

Example VLAN	Purpose / VMware mapping	Core / access configuration
10 MANAGEMENT	ESXi management VMkernel + vCenter administration	Tagged on approved host trunks; authorised administrator sources only.
20 APPLICATIONS	VM port group: PG-APP	Approved application gateway and policy; validate guest IP, DNS and service paths.
30 BACKUP	Backup transport / repository access	Limit to backup components; schedule load and validate restore connectivity.
40 MIGRATION	Dedicated vMotion VMkernel, if enabled	Host-to-host only; validate end-to-end MTU and licensed / compatible migration path.
50 HARDWARE OOB	HPE iLO management interface	Dedicated access port / management network; exclude from VM guest trunks.

UPLINK DESIGN

Use the available NIC inventory to plan active / standby paths. Keep switch and VMware teaming policies consistent; do not assume a cross-switch port channel.

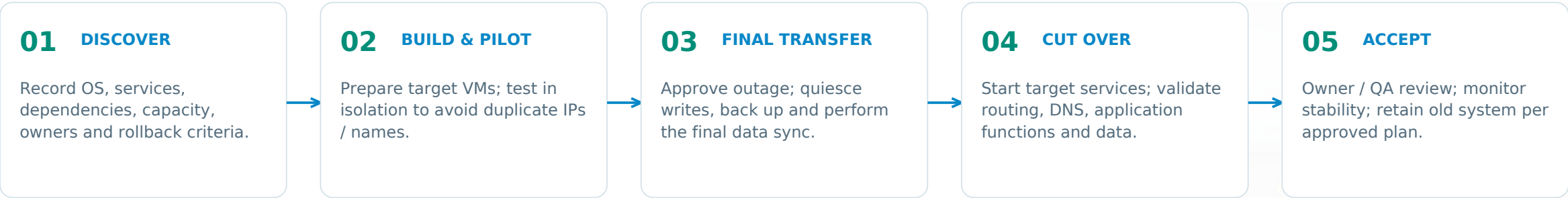
CONFIGURATION CHECKS

Record physical ports, VLAN tags, native-VLAN policy and MTU. Validate DNS / NTP, routing, allowed flows and failover before production cutover.

Standard vSwitch and physical-switch policy must agree. LACP requires a suitable distributed-switch design and confirmed entitlement / hardware support.

Legacy migration | Protect service and data through cutover

Select the migration method per workload: supported P2V for suitable physical systems; compatible cold / live migration for existing VMs; or a fresh VM with application-aware backup and restore. Use a fresh, supported role migration for directory services where required by the application design.



Effort plan	Implementation focus
DAY / EFFORT 1-2	Discovery, backup / restore readiness, hardware and hypervisor build; dependency and network baseline.
DAY / EFFORT 3	Pilot migration and owner testing; record exceptions before the production move.
DAY / EFFORT 4-5	Approved cutover, validation, monitoring and handover. Five man-days are ordered effort, not a guaranteed five-day elapsed completion.

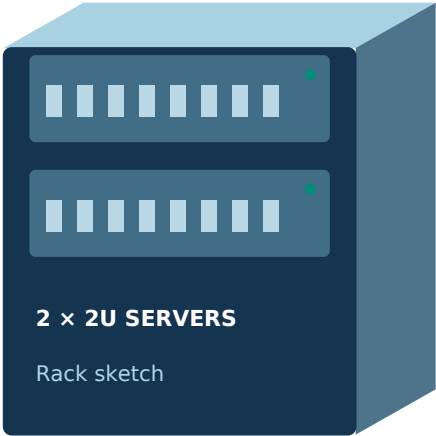
ROLLBACK GATE

If acceptance fails: stop target writes, reconcile any new data, then restore / re-enable the approved source. Prevent both copies from serving production at the same time.

Implementation | HPE build and VMware configuration

Record the exact server BOM and support matrix before installation. The ordered 8SFF chassis format does not identify populated disks or capacity. Use the supported HPE firmware / driver combination for the selected ESXi 8.0 release and retain the approved configuration baseline.

PHYSICAL PLATFORM



2 × 2U SERVERS
Rack sketch

Two DL380 Gen10 servers

Sketch only; rack / disk population is illustrative.

Build stage	Technical work / recorded output
HPE BASELINE	Serial / BOM inventory; firmware, boot mode, controller configuration, hardware health and restricted iLO access.
ESXi 8.0	Compatibility check; supported install media, management IP / DNS / NTP, datastore layout and uplink mapping.
vCENTER 8.0	Appliance deployment; DNS / time consistency, host onboarding, roles, alarms and configuration backup plan.
VM CONFIGURATION	Approved vCPU / RAM / disks, guest OS, VMware Tools and application settings. Size from baseline workload measurements.
LICENSE / CONTROL	Allocate entitlements against physical-core inventory; record keys securely, apply approved access and change controls.

Do not size capacity from the licence count. CPU, RAM, disk layout, NIC speed and the exact switch configuration must come from the asset / build record.
 HA / vMotion / shared storage: enable only after compatible storage, network, capacity and entitlement checks; no SAN or cluster configuration is evidenced in the POs.

Pharma operations | Managed access, changes and recovery

The datacenter supports approved company IT workloads. Its controls should make changes traceable, administration restricted and recovery testable. For systems designated as regulated by the client, application owners and Quality approve the required validation and acceptance evidence; infrastructure access alone does not establish application validation.

ACCESS & OWNERSHIP

Separate viewer, operator and administrator roles. Restrict iLO / ESXi access, use controlled admin accounts and keep credentials in an approved vault.

CHANGE CONTROL

Approve patches and configuration changes. Record request, impact, backup, test, execution and owner / Quality review before closure.

BACKUP & RECOVERY

Use an approved backup target and retention policy. Test guest data / application restore and vCenter recovery; a VM snapshot is not a backup.

Recommended rhythm	Management activity / evidence
DAILY	Check HPE / vCenter alarms, host connectivity, VM services, datastore headroom, time sync and backup job results.
WEEKLY / PLANNED	Review open incidents, capacity trends and patch readiness; investigate repeated faults and failed jobs.
PER APPROVED SCHEDULE	Run restore tests, review access and licences, verify configuration backups and update the asset / VM / port map.

Service path: user / monitoring alert → ticket → triage → approved corrective action → application-owner validation → documented closure.

Operational cadence is a recommended design. Retention, recovery objectives and validation requirements are set by the client; no fixed RPO / RTO or compliance certification is claimed.

Verification | Access confirmed, acceptance defined

Customer access confirmation	Entries	What it establishes
HPE management interfaces	2	Reported working access. The screenshot says “HPE iLOM”; HPE platform terminology is iLO.
VMware ESXi 8.0	2	Reported working host credentials / access.
Windows Server 2022 Standard	2	Two listed Windows systems; VM identities and workload roles are not shown.
vCenter 8.0	1	Reported working vCenter credentials / access.

WINDOWS ENTITLEMENT RECONCILIATION

The PO orders 2 × 16-core Windows Server 2025 Datacenter packs. Verify host physical cores, processor minimums, downgrade / edition rights, activation and CALs against the deployed guests.

VMWARE TERM / CORE RECONCILIATION

The PO lists 32 core units. The separate image repeats 32 cores for years 1–3; it is not 96 simultaneous cores. Confirm redeemed entitlement, dates and host-core allocation.

TECHNICAL CLOSEOUT CHECKS

Host / storage health • trunk and port-group map • backup / restore evidence • migration records • application-owner testing • licence register • configuration backup • signed handover

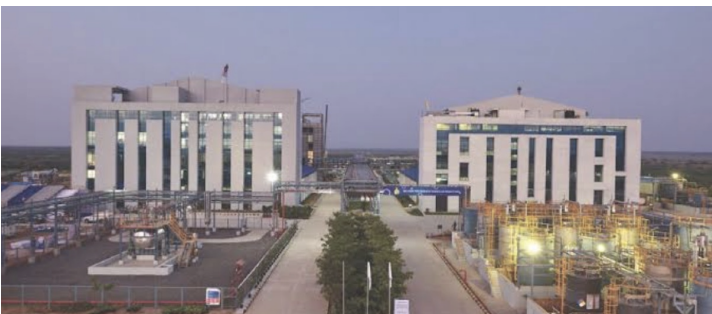
Customer screenshot confirms access, not uptime, storage resilience, completed migration or validated application operation. Those outcomes need the closeout records.

Client & technical handover | PI Health Sciences Limited



PI Health Sciences Limited

Project address in the orders:
 E-969, Biotech Park, RIICO Industrial Area,
 Sitapura, Jaipur 302022, Rajasthan.



CLIENT-SUPPLIED COMPANY FACILITY

Reference	Record
SERVER ORDER	5100006035 22 March 2025 2 × HPE DL380 Gen10 8SFF rack servers.
LICENCE / SERVICE ORDER	5100006083 26 March 2025 Windows Server 2025 Datacenter; VMware vSphere Foundation 8; 5 installation man-days.
SUPPLIED SUPPORTING FILES	Company logo and 2 facility photos; client platform-access message; 32-core / three-year VMware pricing image.

D-TECH | PROJECT SUPPORT

support.dtechindia.com | Login / raise a service request ↗

Support: +91 99989 03042 | support@dtechindia.com



TECHNICAL REFERENCES

- HPE DL380 QuickSpecs
- Broadcom: storage / VM migration
- Microsoft: Windows Server licensing

Source-based technical case study. Architecture and migration sketches explain the delivery design; photos are company context images.